Mario Becerra, MSc.

Data Scientist / Statistician

Location: Leuven, Belgium

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My main interests are data visualization, statistical modeling (particularly Bayesian modeling), and market research.

SKILLS

Statistics and AI

Bayesian statistics, Experimental Design, Multivariate statistics, choice modeling, A/B testing, Machine learning, artificial neural networks, convolutional neural networks, random forests, recommender systems

Programming languages and other technologies

R, Python, SQL, Git, C++, Apache Hive, MATLAB, Excel, PowerPoint

Languages

Spanish: Native English: Native-like competence and accent French: Intermediate (B1) Dutch: Intermediate (A2)

EDUCATION

PhD candidate, KU Leuven, Belgium, (2019 – 2023)

Coursework on statistics and experimental design. I am mostly dedicated to doing full-time research.

Computer Science MSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2016 - 2018)

Relevant coursework: Machine learning, Bayesian generalized linear models, Bayesian multilevel modeling, Big Data, Computer Architecture, Computational Statistics. My thesis was about Bayesian Convolutional Neural Networks.

Applied Mathematics BSc, Instituto Tecnológico Autónomo de México (ITAM), Mexico City (2010 - 2015)

Relevant coursework: Statistical learning, multivariate statistics, Bayesian statistics, generalized linear models, linear programming, numerical optimization, economics, finance, programming, databases. My thesis was about recommender systems.

EMPLOYMENT

Freelance data scientist and programmer

October 2020 - Present

Small freelance projects that include:
- Data visualization.
- Speed-up of slow R code by creating C++ functions on the backend.
- Analysis of Twitter data.
- Choice modeling for market research.

Graduate researcher at KU Leuven, Belgium (Full time)

October 2019 - Present

Research in statistics and experimental design in the Biostatistics group at MeBioS.
I created the *opdesmixr* R package for computing Bayesian D-optimal and I-optimal designs for choice models involving mixtures of ingredients and with the assumption of a multinomial logit model. The package uses C++ under the hood to maximize efficiency. Link to package: https://github.com/mariobecerra/opdesmixr.
I also published a paper with my supervisor, Peter Goos, called *Bayesian I-optimal designs for choice experiments with mixtures* using our package. I have presented my research in five international conferences and seminars so far.
Modeling of preferences and willingness-to-pay for match tickets of supporters of the Belgian national football teams through statistical analysis of a choice experiment.

Data scientist at Banco Azteca, Mexico City (Full time)

September 2018 - September 2019

Data analysis for the marketing and CRM teams of the bank.
I was also in charge of creating and optimizing software for these analyses. Additionally, I managed and helped junior analysts in the area. I also presented results to non-technical stakeholders, such as directors and managers of the bank.
Projects included:
- Visualization and analysis of large customer datasets.
- Models for the prediction of future bank transactions.
- Spatial analysis of customer and branch location data.
- Marketing campaign analysis and causal inference.
- Optimized the sampler of a Bayesian hierarchical model for BTYD analysis suited for large scale data analysis.
- Creation of marketing campaigns based on customer analysis, particularly:
  - Debit customer segmentation based on regularity of deposits and withdrawals.
  - Identification of highly regular remittance receivers from the US and Mexico.
  - Prediction model to identify remittance customers who are willing to accept a bank loan.

EMPLOYMENT

Data scientist at Business Data Evolution, Mexico City (Full time)
September 2017 - September 2018
Developed models both for prediction and for inferential analysis in data science consulting company.
- Analysis of predictive models using Individual Conditional Expectation (ICE) and variable interaction quantification in R.
- Deployed final predictive models in Microsoft Azure VM and Azure Machine Learning.
- Developed Bayesian predictive model for small black and white documents belonging to 4 classes.
- Developed a Bayesian time series demand prediction model for a national grocery store.
- Met with clients to understand their problems, and choose the right model for the data and the clients' needs.
- Presented results to non-technical audiences.

Data scientist at CAD Salud, Mexico City (Full time)
May 2016 - June 2017
Built predictive and inferential statistical models for health projects.
- Performed statistical analysis, exploratory data analysis and developed models in R.
- Created multivariate predictive model for potential diabetes patients in public hospitals.
- Generated spatial mathematical model for vaccines demand in public hospitals in Mexico.
- Aided in the creation of multivariate index to rank public hospitals.

Data Scientist at Grupo Salinas, Mexico City (Full time)
October 2015 - May 2016
Designed and built statistical and machine learning models for predictive and inferential analysis in market research.
- Performed statistical analysis, exploratory data analysis and developed models in R.
- Created data visualization dashboards using RStudio's Shiny.
- Aided in creation of predictive model of loan origination clients in a Mexican bank.

Data analyst at Computer Research and Analysis Laboratory, ITAM, Mexico City (Part time)
June 2014 - June 2015
Natural Language Processing for automatic news classification for media monitoring company.
- Cleansed news articles data for modeling, mainly using R.
- Created different predictive models for automatic news classification in Python using Scikit Learn.

HOBBIES AND INTERESTS

- **Running**: at least once a week for a few kilometers each time
- **Hiking**: going to the forest or mountain with friends or colleagues
- **Cycling**: I like long-distance road cycling with friends and team members, but I also sometimes do mountain biking
- **Skiing:** I enjoy this activity very much, even though I am not the best skier out there
- **Language learning**: I learned French mostly on my own, and I am currently taking Dutch lessons
- **Bicycle maintenance and mechanics:** I like fixing my bikes myself and keeping them in good shape
- **Reading**: mostly non-fiction titles, but also classic and modern novels
- **Traveling**: I like traveling to different countries and cities to experience new cultures
- **Playing guitar**: I have been playing for 18 years, and I am taking lessons to improve my technique on classical guitar
- **Cooking:** being away from my country made me appreciate and learn the recipes of my culture
- **Generative art**: I like combining programming and data visualization to create visually appealing images, particularly maps